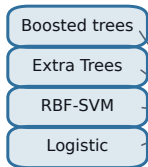


Stage 1 — encoders

Stage 2 — provenance gate

Stage 3 — calibrated decision

screening stream x_s



$$e_s = \frac{1}{K} \sum_k \text{logit } p_s^k$$

clinical stream x_c



$$e_c = \frac{1}{K} \sum_k \text{logit } p_c^k$$

$z \in \mathbb{R}^{12}$
meta-vector

$$a = [\tilde{x}_s; e_s; e_c; e_c - e_s]$$

$$h = \tanh(W_1 a + b_1) \in \mathbb{R}^{16}$$

$$\alpha = \sigma(w_\alpha^\top h + b_\alpha) \in (0, 1)$$

$$l = \alpha e_c + (1 - \alpha) e_s + v^\top \tilde{z} + b_0$$

audit output: $\alpha(x)$ = share of evidence
from clinician-administered items

$$u = \sigma(l)$$

$$p = \Gamma_\theta(u) \quad \text{monotone spline} \\ \Gamma' \geq 0 \text{ by construction } (M = 16 \text{ knots})$$

$$\mathcal{L} = \text{NLL} + \lambda_{\text{cost}} \mathcal{L}_{\text{dec}} + \lambda_{\text{cal}} \mathcal{L}_{\text{ECE}} \\ + \lambda_{\text{ent}} \mathcal{R}_{\text{ent}} + \lambda_2 \|\Theta\|^2$$

$$\hat{y} = \mathbb{1}[p \geq \tau^*] \\ \tau^* = C_{FP} / (C_{FP} + C_{FN}) = 1/6$$